

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

19:39

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# 1. Calculate the survival rate by class

print(data.groupby('Pclass')['Survived'].mean())


# 2. Calculate the survival rate by embarked location

print(data.groupby('Embarked_S')['Survived'].mean())


# 3. Calculate the survival rate by family size

print(data.groupby('FamilySize')['Survived'].mean())


# 4. Calculate the survival rate by being alone

print(data.groupby('IsAlone')['Survived'].mean())


# 5. Get the average fare by class

print(data.groupby('Pclass')['Fare'].mean())


# 6. Get the average age by class

print(data.groupby('Pclass')['Age'].mean())
```

7. Get the average age by survival status

```
print(data.groupby('Survived')['Age'].mean())
```

8. Get the average fare by survival status

```
print(data.groupby('Survived')['Fare'].mean())
```

9. Get the number of survivors by class

```
print(data[data['Survived'] == 1]['Pclass'].value_counts())
```

10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Average time
0.526 s
526.00 ms



Maximum time
0.526 s
526.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 526 ms Debug

Expected output

Actual output

Pclass

Pclass

1...0.629630

1...0.629630

2...0.472826

2...0.472826

3...0.242363

3...0.242363

Name: Survived, dtype: float64

Name: Survived, dtype: float64

Embarked_S

Embarked_S

0...0.506073

0...0.506073

1...0.336957

1...0.336957

Name: Survived, dtype: float64

Name: Survived, dtype: float64

FamilySize

FamilySize

0...0.303538

0...0.303538

1...0.552795

1...0.552795

2...0.578431

2...0.578431

3...0.724138

3...0.724138

4...0.200000

4...0.200000

5...0.136364

5...0.136364

6...0.333333

6...0.333333

7...0.000000

7...0.000000

10...0.000000

10...0.000000

Name: Survived, dtype: float64

Name: Survived, dtype: float64

IsAlone

IsAlone

0...0.505650

0...0.505650

1...0.303538

1...0.303538

Name: Survived, dtype: float64

Name: Survived, dtype: float64

Pclass

Pclass

1...84.154687

1...84.154687

2...20.662183

2...20.662183

3...43.235556

3...43.235556

Activate Windows
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